

Luke Agbetu
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PROFESSIONAL SUMMARY

Lead Data Engineer with 9+ years of experience designing, building, and optimizing scalable data pipelines, cloud-native platforms, and enterprise data solutions across telecom, financial services, and regulated healthcare environments. Deep expertise in Python, SQL, PySpark, Spark, Databricks, and Apache Airflow, with production experience across AWS, Azure, and GCP. Proven record of leading end-to-end data engineering initiatives — from architecture and pipeline design through performance tuning, quality assurance, and stakeholder delivery. Skilled in data lake and warehouse architecture, distributed processing, API integration, schema design, ML pipeline support, and building reliable, observable data systems at scale.

CORE COMPETENCIES

- Enterprise Data Engineering & Platform Development
- Advanced SQL Engineering & Query Optimization
- ETL / ELT Pipeline Design & Automation
- Distributed Data Processing (Batch & Near Real-Time)
- Data Lakehouse & Warehousing Architecture
- Workflow Orchestration (Apache Airflow)
- Cloud Data Engineering (AWS, Azure & GCP)
- Dimensional Data Modeling (Star & Snowflake)
- PostgreSQL / SQL Server / Oracle Database Engineering
- Query Performance Tuning & Execution Plan Analysis
- Production Support, Runbooks & Operational Troubleshooting
- Data Quality, Reliability & Observability
- Cross-Functional Stakeholder Collaboration
- Agile Delivery in Regulated Environments

TECHNICAL SKILLS

Languages & Core Tools

Python, SQL, PySpark, Spark SQL, PL/SQL, PowerShell, Java, Node.js, Express.js, C# (ASP.NET / .NET Core)

Data Engineering & Processing

Apache Spark (Batch & Streaming), Apache Airflow, Databricks, Delta Lake, Kafka, Apache Flink, Hadoop (HDFS, Hive, Pig, Sqoop), dbt, Pandas, NumPy, ETL / ELT Pipelines

Cloud Platforms

AWS (S3, Redshift, Glue, EKS, CloudWatch) | Azure (Data Factory, Databricks, Blob Storage, Azure SQL, HDInsight, Log Analytics, KQL) | GCP (BigQuery, Cloud Storage, Pub/Sub, Dataflow, Dataproc, Cloud Composer)

Databases & Warehousing

Snowflake, Amazon Redshift, Google BigQuery, Oracle Database, PostgreSQL, SQL Server, MongoDB, Teradata, DB2

Data Modeling & Architecture

Star & Snowflake Schemas, SCD (Slowly Changing Dimensions), Medallion Architecture (Bronze / Silver / Gold), Transactional Modeling

DevOps & Infrastructure

Docker, Kubernetes, Terraform, GitHub Actions, CI/CD, Linux

BI & Visualization

Crystal Reports, Power BI, Tableau, Excel (Advanced)

Monitoring, Security & ML

Datadog, SolarWinds, Rapid7 InsightVM, Microsoft Sentinel (SIEM), Scikit-learn, TensorFlow, Keras, Spark MLlib, Vulnerability Management, Compliance, PII Protection

Collaboration & Enterprise Tools

Git, GitHub, Jira, Confluence, Informatica, SSIS, IBM DataStage

RELEVANT EXPERIENCE

Data Engineer

AT&T | January 2025 – Present

- Designed, built, and owned production-grade ETL/ELT pipelines using Python, SQL, Spark, and Apache Airflow, processing 10M+ daily customer and operational records across AT&T's telecom data platform.
- Developed batch and near-real-time Spark pipelines using PySpark and Spark SQL, reducing average processing latency by ~30% through partition tuning, broadcast joins, and execution-plan optimization.
- Implemented data quality checks, schema validation, and anomaly detection across 15+ pipeline DAGs, cutting data reprocessing incidents by ~40% and improving downstream reporting reliability.
- Modeled analytical datasets using star and snowflake schemas, supporting 8+ downstream BI and analytics teams consuming data through Redshift and Snowflake.
- Built CI/CD workflows using GitHub Actions, Docker, and Terraform, reducing pipeline deployment time from hours to under 15 minutes and enabling more reliable production releases.
- Optimized Databricks Delta Lake tables using Z-ordering and VACUUM, reducing query scan times by ~25% on high-volume workloads.
- Containerized 20+ data pipelines using Docker and deployed on Kubernetes (EKS), enabling auto-scaling and fault-tolerant execution across distributed workloads.
- Collaborated with product managers, analysts, solutions architects, and business stakeholders to translate complex reporting and data requirements into production-ready pipelines, models, and analytics-ready datasets.
- Authored technical documentation and operational guidance to improve maintainability, troubleshooting, and production support readiness.

Tools & Technologies: Python, SQL, PySpark, Spark SQL, Apache Airflow, Databricks, Delta Lake, Amazon Redshift, Snowflake, AWS (S3, EKS), Docker, Kubernetes, Terraform, GitHub Actions, Power BI, Git

Data Engineer | Cloud & Analytics Support

Exact Trust, Dallas, TX. | October 2023 – December 2024

- Engineered and operated cloud-based data pipelines across AWS and Azure, supporting analytics and reporting for 6+ internal business units and processing 2TB+ of structured and semi-structured data monthly.
- Built Databricks and Spark-based Medallion pipelines using Delta Lake, reducing data processing time by ~35% and improving data freshness for downstream consumers.
- Used Azure Data Factory, Databricks, and cloud storage services to support ingestion, transformation, and delivery of analytics-ready datasets.

- Integrated Rapid7 vulnerability data into analytics dashboards and compliance workflows, improving incident detection response time by ~20%.
- Automated ingestion, transformation, and reporting workflows using Python and PowerShell, eliminating ~15 hours/week of manual effort across the team.
- Authored SOPs, runbooks, and technical documentation for engineering and operations teams.

Tools & Technologies: Python, PowerShell, Apache Spark, Databricks, Delta Lake, AWS (S3, Glue, Redshift), Azure (Data Factory, Databricks, Blob Storage, Azure SQL), Rapid7 InsightVM, SolarWinds, Power BI, Excel, Git

Data Engineer / SQL Database Developer

DbConn (Hybrid) | Hamburg, Germany | March 2023 – September 2023

- Developed and optimized 30+ complex SQL objects, including stored procedures, views, and table-valued functions for enterprise database environments supporting 5+ business applications.
- Performed advanced query tuning, indexing strategy improvements, and execution-plan analysis, improving query performance by up to 60% on high-frequency reporting and transactional workloads.
- Built backend automation solutions using Python and SQL to streamline customer processes and improve operational efficiency.
- Supported production SQL environments by diagnosing query inefficiencies, troubleshooting database-related issues, and improving stability for downstream applications.
- Contributed to schema and workload optimization efforts across SQL Server-based systems, helping improve maintainability and performance of enterprise data operations.

Tools & Technologies: SQL Server, T-SQL, Crystal Reports, Python, Stored Procedures, Views, Table-Valued Functions, Indexing, Execution Plan Analysis, Git

Earlier Experience (Lead, Software Developer/Technology)

Skye Bank | UBA | Guaranty Trust Bank | Unified Payment Services | Powersoft Integrated Solutions | 2012 – 2022

- Led and supported enterprise technology, digital banking, and data-driven platforms across regulated financial institutions, with responsibility spanning application delivery, core banking integrations, database systems, digital product implementation, and infrastructure modernization.
- Managed and coordinated cross-functional technology units, including database, application development, e-business, network, and digital channels, translating business requirements into scalable technical solutions for banking operations and customer-facing platforms.
- Designed and implemented API-driven integrations using RESTful and SOAP services to connect core banking systems with mobile banking, internet banking, USSD, payment, transaction notification, and real-time reporting platforms.
- Delivered multiple production systems in regulated banking environments, including mobile account opening, real-time payment portals, SMS alert platforms, internet banking services, USSD banking, POS/ATM settlement systems, and government-facing transaction reporting solutions.
- Wrote and supported Oracle stored procedures, SQL-based application logic, and database-driven backend services for mission-critical banking applications.
- Led Oracle core banking upgrade initiatives and supported enterprise database environments, including disaster recovery, data center migration, and business continuity projects.
- Built and supported bespoke web and database applications using Oracle, SQL Server, PostgreSQL, MySQL, C#, Java, and .NET technologies, with hands-on experience in backend development, reporting, and operational support.

- Developed and maintained business and operational reports using Crystal Reports and SQL-based reporting tools, supporting finance, transaction monitoring, and internal business workflows.
- Produced management reporting, project status updates, SOPs, and technical documentation, and worked closely with executive leadership, vendors, and cross-functional business stakeholders on strategic technology initiatives.
- Improved operational efficiency and reduced vendor dependency by delivering in-house solutions that automated workflows, accelerated delivery timelines, and lowered implementation costs.

Tools & Technologies: Oracle Database, PostgreSQL, SQL Server, MySQL, PL/SQL, SQL, Crystal Reports, Stimulsoft, Java, C#, ASP.NET, .NET, Node.js, REST APIs, SOAP, Power BI, Git

EDUCATION

MSc. Computer Science – Towson University, Maryland, United States (Completed)	December 2025
BSc. Computer Science & Mathematics - Fountain University, Osun State, Nigeria	July 2011

CERTIFICATIONS

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- CompTIA Security+ CE Certification
 - AWS Certified Cloud Practitioner Certification
 - AWS Certified Data Engineer - Associate

PUBLICATION

Detecting SARS-CoV-2 in CT scans Using Vision Transformer & GNN — MDPI Algorithms